

Double-stranded genome assembly as a lattice sum, and the inversion-flip group

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Abstract

The two companion notes on double strands ended with a question: the sandpile group $\mathcal{K}(D_k)$ of the reverse-complement double cover counts Eulerian circuits through both strands at once, its deck involution is an anti-automorphism carrying a symmetric linking form and no metabolizer, and what a double-stranded reconstruction count is built from was left open. This note answers it. A circular sequence T is a double-stranded reconstruction of S at word length k when $\text{spec}(T) + \text{spec}(\bar{T}) = \text{spec}(S) + \text{spec}(\bar{S})$; writing m for the arc multiplicities of D_k and $c = \text{spec}(T)$, this says $c + \rho c = m$, and with $f = 2c - m$ it says that f lies in the (-1) -eigenlattice Λ^- of ρ on the circulation lattice of D_k , in the box $|f| \leq m$, in the parity class of m . The count is therefore a *lattice sum*, $\#\text{DS}(S, k) = \sum_f N_{\text{seq}}((m + f)/2)$, each term the BEST count of a multigraph divided by $\prod_a c(a)!$ — and that division, invisible on one strand where it is a constant, is mandatory here because it varies from point to point. The rank of Λ^- is $(|A| - |V| + 2 - \#\text{fix}_A - \#\text{fix}_V)/2$ by the Hopf trace formula applied to an arc-reversing involution; it is *not* the number of inverted repeats, and the guess that it is — made in the planning of this note — is refuted at $k = 12$, where one inverted-repeat pair and one palindrome meet a lattice of rank four. For ϕX174 the sum gives 10 reconstructions at $k = 12$ and 864 at $k = 11$, against $|\mathcal{K}(D_k)| = 33$ and 25 318 912, and an independent enumeration of closed walks with the strand-pairing constraint returns the same numbers point by point. The second half concerns moves. A block $w X \bar{w}$ of T may be replaced by $w \bar{X} w$; every such flip is a flip between a visit of a vertex v of D_k and a later visit of $\rho(v)$, and it changes the spectrum by $-c_B + \rho c_B$. For ϕX174 the 864 reconstructions at $k = 11$ form a *single* component under flips alone — no flip fixes a spectrum, so the same-spectrum repeat exchanges of the single-strand theory are compositions of flips through other classes, of length four, six or eight — and the five inverted-repeat pairs of S do not suffice: six of the 218 spectrum classes need a flip at a pair $\{a, \bar{a}\}$ with a a single-strand repeat, available only after an earlier flip has imported $\bar{a}$. Connectivity is a property of the genome and not a theorem: a sequence with repeats and no inverted repeat has a discrete flip graph.

1 What was asked, and what is true

The single-strand theory of these notes reads the reconstruction count of a circular sequence S at word length k off the critical group of its de Bruijn multidigraph $G_k(S)$ through the BEST theorem [1, 10]. The first companion note [4] passed to the reverse-complement double cover $D_k(S)$, the multidigraph whose arcs are the $(k+1)$ -mer occurrences in S and in $\bar{S}$, and found that everything single-stranded applies to it verbatim but that its involution $\rho(v) = \bar{v}$ reverses

arcs. The second [5] proved that this is fatal for every covering-theoretic route: $P\Delta P = \Delta^\top$, the eigenlattices of ρ are not Δ -stable, the sandpile group $\mathcal{K}(D_k)$ sits in no covering sequence, and what ρ provides instead is a symmetric linking form on $\mathcal{K}(D_k)$ which for ϕ X174 has no metabolizer because $|\mathcal{K}(D_k)|$ is not a square. That note closed with the statement that whatever counts double-stranded reconstructions is not the order of a distinguished subgroup of $\mathcal{K}(D_k)$, and left open what it is.

The answer is that it is not a group order at all. $\mathcal{K}(D_k)$ counts Eulerian circuits of D_k , which are single closed walks through all $2N$ arcs, that is, through both strands. A double-stranded reconstruction is a circular sequence T of length N whose $(k+1)$ -mer spectrum, together with that of $\bar{T}$, is the spectrum of S together with that of $\bar{S}$; it is a closed walk through *half* the arcs, whose ρ -image walks the other half. Writing c for the spectrum of T and m for the multiplicities of D_k , the condition is $c + \rho c = m$. That is a linear condition, and with $f = 2c - m$ it becomes the condition that f be a ρ -anti-invariant circulation in the box $|f| \leq m$ with the parity of m . The reconstruction count is therefore a sum over the lattice points of a polytope of the sequence count of the multigraph at each point (Theorem 6). The rank of the lattice has a closed form (Theorem 9), and the sum, evaluated for ϕ X174, gives 10 and 864 where $|\mathcal{K}(D_k)|$ gives 33 and 25 318 912 (Table 2). An enumeration that never mentions the lattice — closed walks of D_k with the strand-pairing constraint, bucketed by spectrum — returns exactly the connected lattice points, with exactly the predicted count at each (check (E)).

The second half of the note is about moves between reconstructions. If T contains a k -mer w and, later, its reverse complement $\bar{w}$, the block between and including them may be reverse-complemented in place; this is the inversion of classical genome rearrangement, and it is the only move that changes which strand a stretch of T is read from. For ϕ X174 the flip graph on all 864 reconstructions at $k = 11$ is connected (Theorem 14), and the way it is connected is informative: no flip ever preserves the spectrum, so the same-spectrum moves of the single-strand theory — exchanging the two copies of a repeat — arise here only as compositions of flips passing through other spectra, always of even length at least four. This is the de Bruijn-graph shadow of the fact, familiar from sorting by reversals [6, 2], that a transposition is a product of reversals; here each reversal must be flanked by an inverted-repeat pair, and the price is four rather than three.

Three corrections to the specification of this note are recorded in Remark 16: the rank of Λ^- is not the number of inverted repeats; the per-point count must be divided by $\prod c(a)!$; and flip-connectivity is a property of the genome, not a theorem.

Throughout, S is a circular sequence of length N over $\{A, C, G, T\}$, $\bar{S}$ its reverse complement, $\bar{w}$ that of a word w , and a *circular sequence* is a word up to rotation. All computations are exact; the scripts `ds_lattice.py` and `ds_flips.py` reproduce every number below and print pass/fail batteries of nine and four checks.

2 The double cover, its contents, and the spectrum of a reconstruction

Definition 1. $D_k = D_k(S)$ is the multidigraph of [4]: one vertex per k -mer occurring at least twice in S and $\bar{S}$ together, one arc per occurrence of a $(k+1)$ -mer in S or in $\bar{S}$, unitig-contracted so that an arc is a maximal segment between two consecutive repeated k -mers on one strand. An arc is thus a *labelled* object (a strand and a position); its *content* is the segment string from the start of its tail k -mer to the end of its head k -mer; two arcs with the same content are parallel copies. Let V be the vertex set, A the set of contents, $m: A \rightarrow \mathbb{Z}_{\geq 1}$ the number

of arcs of each content, $\ell(a)$ the number of $(k+1)$ -mers of the content a , and $\partial: \mathbb{Z}^A \rightarrow \mathbb{Z}^V$, $\partial a = \text{head}(a) - \text{tail}(a)$. The involution ρ sends a vertex w to $\bar{w}$, a content a to $\bar{a}$, and the arc at position p of one strand to the arc at the mirrored position of the other.

- Lemma 2.** (a) ρ is a fixed-point-free involution on arcs, an involution on contents with $m(\bar{a}) = m(a)$ and $\ell(\bar{a}) = \ell(a)$, and reverses direction: $\partial(\bar{a}) = -\rho \partial a$.
- (b) ρ is free on vertices if and only if no repeated k -mer is its own reverse complement, which is automatic for odd k ; it is free on contents if and only if no content is its own reverse complement, and a content with $m(a) \geq 2$ is a single $(k+1)$ -mer, so this is automatic for even k . On a fixed content $m(a)$ is even.
- (c) D_k has one component if S carries an inverted repeat of length at least k and two otherwise, the two being the S -half and the $\bar{S}$ -half, exchanged by ρ .

Proof. (a) An arc is an occurrence on a strand and ρ exchanges the strands, so it is free on arcs; its image on contents is reverse complementation, which preserves length and, being a bijection on occurrences, preserves multiplicity. The arc $u \rightarrow v$ of content a maps to the arc $\bar{v} \rightarrow \bar{u}$ of content $\bar{a}$. (b) The vertex statement is [5, Prop. 2]. A content with two parallel copies is a repeated segment, so its interior k -mers are repeated; but the interior k -mers of a unitig segment are by construction not repeated, so the segment has no interior k -mer and is a single $(k+1)$ -mer, of odd length when k is even, and no word of odd length equals its reverse complement. If $a = \bar{a}$ then the occurrences of a are paired by ρ with occurrences of a , in pairs, so $m(a)$ is even. (c) is [4, Thm. 3]: all k -mers of S are joined by the closed walk S , all those of $\bar{S}$ by $\bar{S}$, and the two walks meet exactly at k -mers common to S and $\bar{S}$. $\square$

Part (b) has a consequence worth stating: ρ is never free on vertices and contents at once, since palindromic k -mers need even k and palindromic $(k+1)$ -mers need odd k . For $\phi X174$, $k = 12$ has one palindromic repeated 12-mer (TGGAGGCCTCCA at 4483) and no fixed content; $k = 11$ has no fixed vertex and one fixed content, the 12-mer just named, now read as an arc between its two halves. The same word moves from one column of Table 1 to the other as k changes parity.

Definition 3. A double-stranded reconstruction of S at level k is a circular sequence T with

$$\text{spec}(T) + \text{spec}(\bar{T}) = \text{spec}(S) + \text{spec}(\bar{S})$$

as multisets of $(k+1)$ -mers. $\#DS(S, k)$ is their number.

Lemma 4. Let T be a double-stranded reconstruction. Every $(k+1)$ -mer of T is an arc of D_k , T traverses each unitig segment of multiplicity one either once or not at all, and the spectrum of T is therefore a well-defined vector $c = \text{spec}(T) \in \mathbb{Z}_{\geq 0}^A$ with $c + \rho c = m$ and $\partial c = 0$. Moreover $|T| = N$.

Proof. The spectrum identity puts every $(k+1)$ -mer of T in the spectrum of S or $\bar{S}$. A non-repeated k -mer w of D_k has one arc in and one arc out; it occurs once in $S\bar{S}$, hence once in $T\bar{T}$, hence at most once in T , and if it occurs in T then the $(k+1)$ -mers of T through it are its unique in- and out-arcs, so T continues along the unitig in both directions. This makes c constant on unitigs. $c + \rho c = m$ is the spectrum identity read on contents, since $\text{spec}(\bar{T}) = \rho \text{spec}(T)$; $\partial c = 0$ because T is a closed walk. Finally $\sum_a c(a)\ell(a) + \sum_a c(\bar{a})\ell(\bar{a}) = \sum_a m(a)\ell(a) = 2N$ and the two sums on the left are equal. $\square$

3 The lattice sum

Definition 5. $Z_1 = \ker \partial \subset \mathbb{Z}^A$ is the circulation lattice of D_k , ρ acts on it by $(\rho f)(a) = f(\bar{a})$, and Λ^+, Λ^- are its (± 1) -eigenlattices. For a circulation c with $0 \leq c \leq m$ let G_c be the multigraph with $c(a)$ labelled parallel arcs of content a , $t(c)$ its number of spanning arborescences converging on a root (independent of the root, G_c being balanced), $\text{Loc}(c) = \prod_v (d_c^+(v) - 1)!$ over the vertices of positive c -degree, and

$$N_{\text{seq}}(c) = \frac{t(c) \text{Loc}(c)}{\prod_a c(a)!}.$$

For a circular sequence T let $d(T)$ be the number of rotations fixing T .

Theorem 6 (The lattice sum). *The map $T \mapsto \text{spec}(T)$ is a surjection from the double-stranded reconstructions of S at level k onto*

$$P(S, k) = \{c \in \mathbb{Z}^A : 0 \leq c \leq m, c + \rho c = m, \partial c = 0, \text{supp } c \text{ connected}\},$$

and with $f = 2c - m$ the set $P(S, k)$ is the set of $f \in \Lambda^-$ with $|f| \leq m$, $f \equiv m \pmod{2}$ and connected support of $(m + f)/2$. The fibre over c has weighted cardinality $\sum_{\text{spec } T=c} 1/d(T) = N_{\text{seq}}(c)$. Hence

$$\sum_T \frac{1}{d(T)} = \sum_{f \in \Lambda^- \cap \{|f| \leq m, f \equiv m \pmod{2}\}} N_{\text{seq}}\left(\frac{m + f}{2}\right),$$

where the terms with disconnected support vanish, and when every reconstruction is aperiodic the left side is $\#\text{DS}(S, k)$.

Proof. Lemma 4 puts $\text{spec}(T)$ in $P(S, k)$; the support is connected because T is one closed walk. Conversely a point $c \in P(S, k)$ defines a balanced connected multigraph G_c , which has an Eulerian circuit, and reading the circuit gives a circular sequence T with $\text{spec}(T) = c$ and, by $c + \rho c = m$, with the spectrum identity. The substitution $f = 2c - m$ is a bijection between $\{0 \leq c \leq m\}$ and $\{|f| \leq m, f \equiv m \pmod{2}\}$; $\partial c = 0$ is $\partial f = 0$ since $\partial m = 0$; and $c + \rho c = m$ is $f + \rho f = 0$.

For the fibre, the BEST theorem [1, 10] counts the Eulerian circuits of G_c , as cyclic sequences of its labelled arcs, by $t(c) \text{Loc}(c)$. Forgetting the labels sends a circuit to a circular sequence with spectrum c . Over a fixed T , a circuit is an assignment of the $c(a)$ labelled copies of each content to the $c(a)$ occurrences of that content around T , $\prod_a c(a)!$ assignments in all, and two assignments give the same cyclic labelled sequence exactly when they differ by a rotation of T onto itself. The rotation group of T , of order $d(T)$, acts freely on assignments, so the fibre has $\prod_a c(a)!/d(T)$ elements, and summing over T gives $t(c) \text{Loc}(c) = \prod_a c(a)! \sum_T 1/d(T)$. If the support of c is disconnected then G_c has no spanning arborescence and $t(c) = 0$. $\square$

Remark 7 (The normalisation is not optional). On one strand there is a single lattice point, $c = m$, and $\prod_a m(a)!$ is a constant; the single-strand notes count in the *arc-rooted* convention $t(m) \text{Loc}(m)$ and never had to divide. Across the lattice the divisor $\prod_a c(a)!$ changes from point to point, and the un-normalised sum $\sum_f t(c_f) \text{Loc}(c_f)$ is $\sum_T \prod_a c_T(a)!/d(T)$, which counts each reconstruction with a weight of its own. The two agree whenever $m \leq 1$, which is the case at $k = 12$ for $\phi X174$, and disagree as soon as a content has two copies: at $k = 11$ the un-normalised sum is 2756 and the count is 864 (check (N)). The single-strand number quoted for $\phi X174$ at $k = 11$ in [3], $|\mathcal{K}(G_{11})| \text{Loc} = 132$, is the arc-rooted count; the number of circular sequences with the spectrum of S is $132/4 = 33$, and 33 is what appears at the point $c = \text{spec}(S)$ of Table 2.

Remark 8 (Where the linking form went). The symmetric linking form of [5, Rem. 5] does not enter Theorem 6, and the guess that a double-stranded count is a Gauss sum or an isotropic count on $\mathcal{K}(D_k) = \mathbb{Z}/3 \oplus \mathbb{Z}/11$ any non-degenerate quadratic form has a single isotropic vector and a Gauss sum of modulus $\sqrt{33}$, while $\#DS(S, 12) = 10$. What survives of the form is only the anti-invariance $f + \rho f = 0$, which is the linear shadow of ρ being a duality rather than a symmetry.

Theorem 9 (Rank of the anti-invariant lattice). *Let $\#\text{fix}_V$ be the number of ρ -fixed vertices and $\#\text{fix}_A$ the number of ρ -fixed contents of $D_k(S)$. Then*

$$\text{rank } \Lambda^- = \frac{|A| - |V| + 2 - \#\text{fix}_A - \#\text{fix}_V}{2}, \quad \text{rank } \Lambda^+ = \frac{|A| - |V| - 2 + \#\text{fix}_A + \#\text{fix}_V}{2} + b_0,$$

where $b_0 \in \{1, 2\}$ is the number of components. The expression does not depend on the compaction: $|A| - |V|$ and $\#\text{fix}_A - \#\text{fix}_V$ are the same for the unitig-contracted graph and for the full de Bruijn graph of $S\bar{S}$.

Proof. Let $C_1 = \mathbb{Z}^A$, $C_0 = \mathbb{Z}^V$ with the boundary ∂ , and let P_A, P_V be the permutation matrices of ρ on contents and vertices. By Lemma 2(a), $\partial P_A = -P_V \partial$, so $\varphi = (-P_A, P_V)$ is a chain endomorphism of the cellular chain complex of D_k , and the Hopf trace formula gives

$$\text{tr}(\varphi_*|H_1) - \text{tr}(\varphi_*|H_0) = \text{tr}(-P_A|C_1) - \text{tr}(P_V|C_0) = -\#\text{fix}_A - \#\text{fix}_V.$$

Here $H_1 = Z_1$ with $\varphi_* = -\rho$, and $H_0 = \mathbb{Z}^{b_0}$ with φ_* the permutation of components induced by ρ , whose trace is the number of ρ -fixed components: 1 when $b_0 = 1$, and 0 when $b_0 = 2$ since then ρ exchanges the halves (Lemma 2(c)). Hence $\text{tr}(\rho|Z_1) = \#\text{fix}_A + \#\text{fix}_V - \text{tr}(\varphi_*|H_0)$. Over $\mathbb{Q}$ the involution ρ is diagonalisable with eigenvalues ± 1 , so $\text{rank } \Lambda^+ - \text{rank } \Lambda^- = \text{tr}(\rho|Z_1)$ while $\text{rank } \Lambda^+ + \text{rank } \Lambda^- = \text{rank } Z_1 = |A| - |V| + b_0$. Solving,

$$\text{rank } \Lambda^- = \frac{1}{2}(|A| - |V| + b_0 + \text{tr}(\varphi_*|H_0) - \#\text{fix}_A - \#\text{fix}_V),$$

and $b_0 + \text{tr}(\varphi_*|H_0) = 2$ in both cases. For the last sentence: unitig contraction is a ρ -equivariant homotopy equivalence, so $\text{rank } Z_1$ and the Lefschetz number $\text{tr}(\varphi_*|H_1) - \text{tr}(\varphi_*|H_0)$ are unchanged, and the latter is $-\#\text{fix}_A - \#\text{fix}_V$ in either cellular structure. $\square$

Corollary 10. *For even k , $\text{rank } \Lambda^- = (|A| - |V| + 2 - \#\{\text{palindromic repeated } k\text{-mers}\})/2$; for odd k , $\text{rank } \Lambda^- = (|A| - |V| + 2 - \#\{\text{palindromic } (k+1)\text{-mers of } S\bar{S}\})/2$. If S has no inverted repeat of length $\geq k$ then $\Lambda^- \cong Z_1(G_k(S))$.*

Proof. The first two statements are Theorem 9 with Lemma 2(b). If D_k is disconnected then $Z_1(D_k) = Z_1(G_k(S)) \oplus Z_1(G_k(\bar{S}))$ with ρ exchanging the summands, and $f \mapsto (f, -\rho f)$ identifies $Z_1(G_k(S))$ with Λ^- . $\square$

Remark 11 (Not the number of inverted repeats). It is tempting to read $\text{rank } \Lambda^-$ as the number of independent inversions available to the genome, and both the specification of this note and its planning did. Table 1 refutes it. At $k = 12$, ϕX174 carries exactly one inverted-repeat pair of length ≥ 12 (ATTAAGCTCATT at 441 against its reverse complement at 1448) and one palindromic repeated 12-mer, and $\text{rank } \Lambda^- = 4$. At the other extreme, Corollary 10 says that a genome with repeats but no inverted repeat has Λ^- as large as the whole circulation lattice of $G_k(S)$, while only two of its box points — $\text{spec}(S)$ and $\text{spec}(\bar{S})$ — have connected support. The number that does count inversion classes is the number of connected box points, 8 and 218 in Table 2, and it has no closed form we know of.

k	$ V $	$\#\text{fix}_V$	$ A $	$\#\text{fix}_A$	$\sum m$	b_0	$\text{rank } Z_1$	$\text{rank } \Lambda^-$
13	0	0	0	0	0	—	—	0
12	7	1	14	0	14	1	8	4
11	30	0	53	1	60	1	24	12
10	125	3	222	0	252	1	98	48
9	396	0	685	3	812	1	290	144

Table 1: ϕX174 (NC_001422.1, 5386 bp). $\text{rank } \Lambda^-$ is computed as the corank of the balance matrix on orbit representatives, modulo two large primes, and agrees with Theorem 9 in every row. At $k = 13$ no k -mer is repeated on either strand and D_k is the pair of disjoint cycles $S, \bar{S}$. Check (L).

k	box	conn.	lattice	brute	un-norm.	one strand	$ \mathcal{K}(D_k) $
12	10	8	10	10	10	4	33
11	528	218	864	864	2756	66	25 318 912
10	rank 48: beyond the enumerator					8 610 708 632	$3.42 \cdot 10^{35}$

Table 2: The lattice sum for ϕX174 . “Box” is $|\Lambda^- \cap \{f \mid f \equiv m\}|$; “conn.” the number of points with $N_{\text{seq}} > 0$; “lattice” is $\#\text{DS}$ from Theorem 6; “brute” the number of closed walks of D_k with usage $+\rho$ usage $= m$, enumerated by backtracking without reference to the lattice; “un-norm.” is $\sum_f t(c_f) \text{Loc}(c_f)$; “one strand” is $N_{\text{seq}}(S) + N_{\text{seq}}(\bar{S})$. $\text{Loc}(D_k) = 1$ at $k = 12, 11$, so $|\mathcal{K}(D_k)|$ is also the Eulerian circuit count of the double cover. Checks (B), (S), (E), (N).

What the lattice points are. At $k = 12$ the eight connected points are $\text{spec}(S)$ and $\text{spec}(\bar{S})$, each with $N_{\text{seq}} = 2$ (the interleaved two-copy repeat of the single-strand theory), and six spectra with $N_{\text{seq}} = 1$. One of the six is the inversion of $S[441..1459]$ between the inverted-repeat pair. The other five mix that inversion with the repeat exchange and with hairpin turns at the palindrome: for instance the point that drops the two segments $S[264..452]$ and $S[1448..2771]$ of S and takes their reverse complements is the sequence

$$\begin{array}{c} S[441..457] \ S[3205..4483] \ S[4483..264+N] \ S[2760..3205] \ S[457..1448] \\ \overline{S[264..441]} \ \overline{S[1448..2760]}, \end{array}$$

a rearrangement of the whole genome in which two blocks are read from the other strand. At $k = 11$ the 218 connected points carry N_{seq} values 33 (twice), 18, 16 (twice each), 15 (four times), 11 (six), 10 (two), 9 (six), 6 (22), 5 (20), 4 (six), 3 (36), 2 (56) and 1 (54), summing to 864. Every one of the 874 reconstructions at the two levels is aperiodic, so $d(T) = 1$ throughout and the weighted count is the count (check (A)).

The brute force. Check (E) enumerates, by depth-first search on the content multigraph, every closed walk whose usage u satisfies $u(a) + u(\bar{a}) \leq m(a)$ along the way and $= m(a)$ at the end — the walk has exactly $\sum m/2$ arcs — rooted at its smallest content a_0 and weighted $1/u(a_0)$; the search never sees Λ^- . It returns 90 nodes and 8 spectra at $k = 12$, 45 061 nodes and 218 spectra at $k = 11$; the spectra found are exactly the connected box points, and the count at each spectrum is exactly N_{seq} .

4 The inversion flip

Definition 12. Let T be a double-stranded reconstruction, read as the closed walk $a_0 a_1 \cdots a_{L-1}$ of arcs of D_k with vertex visits $v_i = \text{tail}(a_i)$. A *flip* at a pair of visits $i \neq j$ with $v_j = \rho(v_i)$ replaces the block $a_i \cdots a_{j-1}$ (indices cyclic) by $\bar{a}_{j-1} \cdots \bar{a}_i$:

$$T' = a_0 \cdots a_{i-1} \bar{a}_{j-1} \cdots \bar{a}_i a_j \cdots a_{L-1}.$$

The *flip graph* has the reconstructions as vertices and the flips as edges.

Proposition 13. (a) T' is a double-stranded reconstruction, and every replacement of a block $w X \bar{w}$ of T by $w \bar{X} \bar{w}$, for a k -mer w and a later occurrence of $\bar{w}$, is a flip in the sense of Definition 12.

- (b) $\text{spec}(T') = \text{spec}(T) - c_B + \rho c_B$, where c_B is the usage of the block. A flip preserves the spectrum if and only if $c_B = \rho c_B$.
- (c) Flipping the complementary block gives $\bar{T}'$; flipping the whole cycle at a palindromic visit gives $\bar{T}$.
- (d) If S carries no inverted repeat of length $\geq k$, no reconstruction admits a flip: the flip graph is discrete, with $2N_{\text{seq}}(S)$ vertices.

Proof. (a) The block $a_i \cdots a_{j-1}$ runs from v_i to $v_j = \rho(v_i)$; its reverse-complement image $\bar{a}_{j-1} \cdots \bar{a}_i$ runs from $\rho(v_j) = v_i$ to $\rho(v_i) = v_j$, so T' is a closed walk, and its spectrum is $\text{spec}(T) - c_B + \rho c_B$, which still satisfies $c + \rho c = m$. For the second statement: the block $w X \bar{w}$ has reverse complement $\bar{w} \bar{X} \bar{w} = w \bar{X} \bar{w}$, so the endpoints are unchanged and the interior $(k+1)$ -mers are replaced by their reverse complements, which lie in $\text{spec}(\bar{T})$ and hence in the double-stranded spectrum. It remains to see that w is a vertex of D_k : w occurs in T and $\bar{w}$ occurs in T , so w occurs in T and in $\bar{T}$, so w occurs at least twice in $T\bar{T}$, hence at least twice in $S\bar{S}$ by the spectrum identity, hence is a repeated k -mer, i.e. a vertex. (b) is the computation in (a). (c) The complementary block runs from v_j to $v_i = \rho(v_j)$, and flipping it yields the walk $\bar{a}_{i-1} \cdots \bar{a}_0 \bar{a}_{L-1} \cdots \bar{a}_j a_j \cdots$, which as a circular sequence is the reverse complement of T' ; taking $i = j$ at a palindromic vertex flips everything. (d) By Lemma 2(c) D_k is then disconnected, T lies in one half, and $\rho(v_i)$ lies in the other for every visit, so no pair of visits qualifies; the reconstructions are those of S and of $\bar{S}$. $\square$

Part (d) says that connectivity of the flip graph is not a theorem: a genome with a multi-copy repeat and no inverted repeat has $2N_{\text{seq}}(S) > 2$ reconstructions and no move between any of them. What can be asked is whether a *given* genome is flip-connected, and the answer for ϕX174 is yes, in a strong form.

Theorem 14 (ϕX174 , computed). *At $k = 12$ and $k = 11$:*

- (a) *the flip graph on all reconstructions (10, resp. 864 vertices; 16, resp. 5140 edges) is connected, without the move $T \mapsto \bar{T}$;*
- (b) *no flip preserves the spectrum: $c_B \neq \rho c_B$ for every block of every reconstruction; consequently the quotient graph on spectrum classes (8, resp. 218 vertices) is connected, and every class is at flip distance at most 3, resp. 8, from $\text{spec}(S)$ (histogram 1, 2, 3, 2, resp. 1, 7, 16, 38, 56, 52, 34, 12, 2);*

- (c) *two reconstructions with the same spectrum are joined by flip paths of even length at least four: at $k = 11$ the 6776 ordered same-spectrum pairs have flip distances 4 (3932 pairs), 6 (2556) and 8 (288); at $k = 12$ all four have distance 4;*
- (d) *the vertex pairs $\{v, \rho v\}$ realising flips number 3, resp. 14, of which 1, resp. 5, are the inverted-repeat pairs and palindromes of S itself; restricted to those, flips reach 6 of 8, resp. 212 of 218, classes. The others are pairs $\{a, \bar{a}\}$ with a a repeated k -mer of one strand — **AAACGGCAGAA** against **TTCTGCCGTTT**, for instance — which become flip sites only after an earlier flip has brought $\bar{a}$ into the reconstruction.*

The theorem is the transcript of `ds_flips.py`: all reconstructions are taken from the brute force of check (E), all flips are generated from Definition 12, and each flip is verified to land in the reconstruction set (13928 instances at $k = 11$).

Remark 15 (Transpositions as products of flips). Part (c) is the de Bruijn-graph form of a classical fact. The two reconstructions of $\text{spec}(S)$ at $k = 12$ differ by exchanging the two copies of a repeat, which as a rearrangement of S is a transposition of the blocks between the copies. In sorting by reversals [2, 6] a transposition is a product of three reversals. Here a reversal is available only when it is flanked by a vertex and its reverse complement, and the shortest product that realises the exchange has four factors; the distances in (c) are all even, which we have not explained. Part (d) shows the mechanism: the exchange needs a flip at $\{a, \bar{a}\}$ with a a single-strand repeat, and the only way to bring $\bar{a}$ into T is to flip first at an inverted-repeat pair of S , then at $\{a, \bar{a}\}$, and then to undo the first flip.

Remark 16 (Corrections to the specification). Three statements in the specification of this note, one of them originating in its planning, are not correct as stated. *First*, $\text{rank } \Lambda^-$ is not the number of independent inverted repeats of length $\geq k$ (Remark 11); it is given by Theorem 9. *Second*, the per-point term is not $t(c_f) \prod (d^+ - 1)!$ but that quantity divided by $\prod_a c_f(a)!$ (Remark 7); the un-normalised sum is 2756 where the count is 864. *Third*, “the flip graph is connected” and “a same-spectrum transposition is a composition of inversions” are properties of ϕX174 at $k = 12, 11$, established by computation, and not theorems: Proposition 13(d) gives the general obstruction, and the composition, where it exists, costs four flips rather than the three reversals of the unconstrained model.

5 Scope, provenance, and what is open

Scope. Lemma 2, Lemma 4, Theorem 6, Theorem 9, Corollary 10 and Proposition 13 are proved in full generality for a single circular sequence. Theorem 14 is a computation for one genome at two word lengths. The lattice sum is evaluated at $k = 12$ and $k = 11$; at $k = 10$ the lattice has rank 48 and the capacity-pruned backtracking enumerator of `ds_lattice.py` produces no point in ten minutes, so only the rank row of Table 1 is reported there. The count is a count of circular sequences, each weighted by $1/d(T)$; in the data $d(T) = 1$ throughout.

Provenance. The BEST theorem is [1, 10]; its use to count genome reconstructions is [8]; the bidirected-graph model of double-stranded assembly, in which a double-stranded reconstruction is a walk in a bidirected de Bruijn graph, is [9] after [7], and those papers ask for existence and for one solution, not for the count. Sorting by reversals and the transposition-as-three-reversals identity are [2, 6]. The double cover D_k , its balance and its connectivity criterion are [4]; the

anti-automorphism theorem and the symmetric linking form are [5]; the single-strand counting convention is [3]. The Hopf trace formula is standard. We are not aware of prior work expressing a double-stranded reconstruction count as a sum over an eigenlattice of the strand involution, or of prior work on the flip graph of a de Bruijn graph; a targeted search found none.

Open. *First*, the number of connected box points — the number of spectrum classes, or equivalently the size of the orbit of $\text{spec}(S)$ under the flip pseudo-group — has no formula; 8 and 218 are all that is known. *Second*, the genome-scale question: Theorem 9 evaluates the rank of Λ^- for *E. coli* in one line from the double cover of [4], but the sum is not enumerable there, and what is wanted is a transfer-matrix or generating-function evaluation of $\sum_f N_{\text{seq}}(c_f)$ along the genome, or an Ehrhart-type asymptotic in k . *Third*, the parity of same-spectrum flip distances in Theorem 14(c). *Fourth*, the same lattice with the involution removed and m split into p parts is the phasing problem for a polyploid, where the symmetric group on the parts acts on the decompositions rather than on the graph; that is the subject of the next note.

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